



Modelling Financial Markets

Seminar 6: Multivariate Volatility and Dependence — Hands-on



Daniel Traian PELE

Drd. Antoaneta Amza, antoaneta.amza@csie.ase.ro

Bucharest University of Economic Studies

IDA Institute Digital Assets

Blockchain Research Center

AI4EFin Artificial Intelligence for Energy Finance

Romanian Academy, Institute for Economic Forecasting

MSCA Digital Finance

Acronyms used in this material

- ▣ AI: Artificial Intelligence
- ▣ AIC: Akaike Information Criterion
- ▣ BAC: Bank of America Corporation (NYSE ticker)
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BUX: Budapesti Értéktőzsde index (the Budapest Stock Exchange index)
- ▣ CCC: Constant Conditional Correlation (model)
- ▣ CML: Canonical Maximum Likelihood
- ▣ COVID-19: COronaVirus Disease 2019
- ▣ DAX: Deutscher Aktienindex (the German stock index (Frankfurt Stock Exchange))
- ▣ DCC: Dynamic Conditional Correlation
- ▣ DCC-GARCH: Dynamic Conditional Correlation GARCH (model)
- ▣ EU: European Union
- ▣ EWMA: Exponentially Weighted Moving Average
- ▣ FRED: Federal Reserve Economic Data
- ▣ FTSE: Financial Times Stock Exchange (FTSE Russell, index provider)
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ GAS: Generalised Autoregressive Score (model)
- ▣ JPM: JPMorgan Chase & Co. (NYSE ticker)
- ▣ PX: Prague Stock Exchange index (index Burzy cenných papírů Praha)
- ▣ STOXX50E: Euro Stoxx 50 index (symbol of the data file)
- ▣ US: United States
- ▣ WIG20: Warszawski Indeks Giełdowy 20 (the Warsaw Stock Exchange index of the 20 largest companies)

Outline

Introduction

Part A: Derivations

Part B: Estimation and Inference

Part C: Open Question

Wrap-up

References

Seminar Objectives

By the end of this seminar you can

- ▣ Compute tail dependence and copula parameters step by step
- ▣ Estimate a DCC model, test it against constant correlation and interpret its persistence
- ▣ Fit and test copulas on pseudo-observations with a parametric bootstrap
- ▣ Separate contagion from the volatility bias (Forbes–Rigobon)

- ▣ Background: Lecture 6; textbooks [Franke, Härdle & Hafner \(2019\)](#) and [Nelsen \(2006\)](#)

Structure

- ▣ **Part A:** six derivations on paper (A1–A6)
- ▣ **Part B:** six computational exercises with inference (B1–B6)
- ▣ **Part C:** one open question, a seed for the team project (C1), a critique of an AI answer (C2) and a project seed from our research (C3)

How to Work in This Seminar

- ▣ **[Solved]** problems show the full worked solution; **[Proposed]** problems show only the task: their solutions are discussed in class
- ▣ Every proposed problem names its **model**: the solved problem that uses the same technique
 - ▶ Every Part B exercise ends with interpretation questions: ask what would change the conclusion
- ▣ Report every estimate with its uncertainty: standard error, bootstrap interval or p -value
- ▣ Ask each time whether the standard method can fail here: time zones, boundary parameters, volatility changes, many observations

Solved and Proposed Problems

Solved: worked examples in this deck

- Part A: A1 (deriving the tail dependence of the t copula), A2 (Kendall's tau, copula parameters and their standard errors), A3 (Forbes–Rigobon adjustment with an estimated δ)
- Part B: B1 (DCC on SPY and TLT, break at an unknown date, two-step standard errors), B2 (Gaussian versus t copula, sandwich standard errors)

Proposed: you solve them

- Part A: A4 (model: A3), A5 (model: B1), A6 (model: A2)
- Part B: B3 (model: B2, A2), B4 (model: A3), B5 (model: B1), B6 (model: B2)
- Part C: C1, open question and project seed; C2, critique of an AI answer (model: A1, A2); C3, energy prices and EU stock markets (model: B1, B2, C1)

Setup: Python Environment and Data

-  **Open in Colab:** `chapter6_seminar_notebook.ipynb`
- Google Colab (recommended) or local Python ≥ 3.10 with `numpy`, `pandas`, `scipy`, `arch`, `matplotlib`
- Series: SPY, TLT, S&P 500, Euro Stoxx 50, BET, Bitcoin; daily closes up to 18 September 2026
- Joint analyses: prices are first joined on common trading days, then returns are computed
- Weekly returns: last common close of each week (Friday); 2-day returns: $\ln P_t - \ln P_{t-2}$

Formulas Used Today

- DCC: $Q_t = (1 - a - b)\bar{Q} + a\varepsilon_{t-1}\varepsilon_{t-1}^\top + bQ_{t-1}$, $R_t = \text{diag}(Q_t)^{-1/2}Q_t\text{diag}(Q_t)^{-1/2}$ Engle (2002)
- Forbes–Rigobon: $\delta = \sigma_{x,c}^2/\sigma_{x,calm}^2 - 1$, $\rho^* = \rho_c/\sqrt{1 + \delta(1 - \rho_c^2)}$ Forbes & Rigobon (2002)
- Pseudo-observations: $\hat{u}_{it} = \text{rank}(x_{it})/(T + 1)$ Genest, Ghoudi & Rivest (1995)
- Tail dependence: $\lambda_L = \lim_{q \downarrow 0} C(q, q)/q$; t copula: $\lambda = 2t_{\nu+1}(-\sqrt{(\nu+1)(1-\rho)/(1+\rho)})$ Demarta & McNeil (2005)
- Kendall's tau: Gaussian/t $\tau = \frac{2}{\pi} \arcsin \rho$; Clayton $\theta/(\theta + 2)$; Gumbel $1 - 1/\theta$
- Goodness of fit: Cramér–von Mises distance of the Rosenblatt pairs $(u, h(v | u))$ from independence, parametric bootstrap Genest, Rémillard & Beaudoin (2009)

A1: Tail Dependence of the t Copula: Task (1/2) [Solved]

Context and data

- No data are needed for (a)–(c); (d) and (e) use $\rho = 0.5$ and $\nu \in \{3, 4, 10, 30\}$
- Definition: $\lambda_L = \lim_{q \downarrow 0} C(q, q)/q$; the copula is exchangeable, $C(u, v) = C(v, u)$
- Fact: for a bivariate Student t with correlation ρ and ν degrees of freedom, given $X_1 = x$ the ratio $(X_2 - \rho x) / \sqrt{(\nu + x^2)(1 - \rho^2)/(\nu + 1)}$ follows $t_{\nu+1}$

What to do

- (a) Use L'Hôpital's rule to show that $\lambda_L = 2 \lim_{q \downarrow 0} P(V \leq q \mid U = q)$
- (b) Substitute the conditional distribution of the t copula and let $x = t_{\nu}^{-1}(q) \rightarrow -\infty$ to derive $\lambda = 2 t_{\nu+1}(-\sqrt{(\nu + 1)(1 - \rho)/(1 + \rho)})$
- (c) Repeat the argument for the Gaussian copula and show that $\lambda = 0$ whenever $|\rho| < 1$

A1: Tail Dependence of the t Copula: Task (2/2) [Solved]

What to do

- (d) Evaluate λ of the t copula with $\rho = 0.5$ for each ν
- (e) Compute the finite-level probability $P(V \leq q \mid U \leq q) = C(q, q)/q$ at $q = 5\%$ and $q = 0.1\%$ for the Gaussian copula and for the t copula with $\nu = 4$, from the bivariate distribution functions

What to report and interpret

- The derivation of (a)–(c) as a chain of equalities, each step justified in one sentence
- A table of λ by ν and the four finite-level probabilities of (e)
- Interpretation: why can a Gaussian risk model calibrated at the 5% level miss joint crashes at the 0.1% level?

A1: Tail Dependence of the t Copula: Solution (1/2) [Solved]

Solution

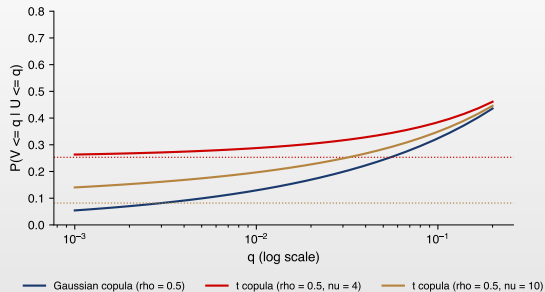
- (a) Both numerator and denominator of $C(q, q)/q$ tend to 0, so by L'Hôpital $\lambda_L = \lim_{q \downarrow 0} \frac{d}{dq} C(q, q)$
- By the chain rule $\frac{d}{dq} C(q, q) = \partial_1 C(q, q) + \partial_2 C(q, q)$, and $\partial_1 C(u, v) = P(V \leq v \mid U = u)$; exchangeability makes the two terms equal, so $\lambda_L = 2 \lim_{q \downarrow 0} P(V \leq q \mid U = q)$
- (b) With $x = t_\nu^{-1}(q)$: $P(V \leq q \mid U = q) = P(X_2 \leq x \mid X_1 = x) = t_{\nu+1} \left(\frac{(1-\rho)x}{\sqrt{(\nu+x^2)(1-\rho^2)/(\nu+1)}} \right)$
- As $q \downarrow 0$, $x \rightarrow -\infty$ and $x/\sqrt{\nu+x^2} \rightarrow -1$; since $(1-\rho)/\sqrt{1-\rho^2} = \sqrt{(1-\rho)/(1+\rho)}$, the argument tends to $-\sqrt{(\nu+1)(1-\rho)/(1+\rho)}$, which gives the formula
- (c) Gaussian: $P(V \leq q \mid U = q) = \Phi(x\sqrt{(1-\rho)/(1+\rho)})$ with $x = \Phi^{-1}(q) \rightarrow -\infty$, so the probability tends to 0 and $\lambda = 0$
- Interpretation: in the t case the conditional scale grows with $|x|$, so the standardised argument stays bounded; in the Gaussian case the scale is fixed and the argument diverges

A1: Tail Dependence of the t Copula: Solution (2/2) [Solved]

Solution

- (d) Formula: $\lambda = 2 t_{\nu+1}(-a_\nu)$ with $a_\nu = \sqrt{(\nu+1)(1-\rho)/(1+\rho)} = \sqrt{(\nu+1) \cdot 0.5/1.5}$
- Substituted values: $a_3 = 1.155$, $a_4 = 1.291$, $a_{10} = 1.915$, $a_{30} = 3.215$
- Intermediate result for $\nu = 4$: $t_5(-1.291) = 0.1266$, hence $\lambda = 2 \times 0.1266 = 0.253$
- All values: $\lambda = 0.313$ ($\nu = 3$), 0.253 ($\nu = 4$), 0.082 ($\nu = 10$), 0.003 ($\nu = 30$): tail dependence vanishes as ν grows
- (e) Gaussian: 0.244 at 5% and 0.054 at 0.1%; t with $\nu = 4$: 0.339 and 0.263
- Interpretation: at 5% the two models look alike, but at 0.1% the Gaussian probability has fallen to a fraction of the t one; a model calibrated on moderate losses cannot reveal this

A1: The Picture [Solved]



- Exact $P(V \leq q | U \leq q)$ from the bivariate Normal and t distribution functions; dotted lines: the limits λ
- Read the chart from right to left: as q falls, the Gaussian curve heads to 0 and the t curves flatten at their limits λ

A2: Kendall's Tau, Copula Parameters and Their Precision: Task (1/2) [Solved]

Context and data

- Data: weekly log returns of the S&P 500 (GSPC.INDX) and the Euro Stoxx 50 (STOXX50E.INDX), Friday closes on common trading days, 2000–2026, $n = 1393$
- Pseudo-observations $\hat{u}_i = \text{rank} / (n + 1)$; weekly pairs treated as i.i.d.
- Sample values: Kendall's $\hat{\tau} = 0.564$, Spearman's $\hat{\rho}_S = 0.746$, Pearson $\hat{\rho} = 0.781$

What to do

- (a) Invert $\hat{\tau}$ for the Gaussian ($\rho = \sin(\pi\tau/2)$), Clayton ($\theta = 2\tau/(1 - \tau)$), Gumbel ($\theta = 1/(1 - \tau)$) and Frank (solve $1 - \frac{4}{\theta}[1 - D_1(\theta)] = \hat{\tau}$ numerically) copulas
- (b) Compute the tail dependence implied by each fitted copula
- (c) Compute Spearman's rho implied by the Gaussian copula, $\rho_S = \frac{6}{\pi} \arcsin(\rho/2)$, and compare it with the sample value

A2: Kendall's Tau, Copula Parameters and Their Precision: Task (2/2) [Solved]

What to do

- (d) From the Hoeffding projection $W_i = 2\hat{C}_n(\hat{U}_i, \hat{V}_i) - \hat{U}_i - \hat{V}_i$ (sample sd 0.1239), show that $se(\hat{\tau}) = 4 \text{sd}(W)/\sqrt{n}$ and compute it; then the delta-method se of $\hat{\theta}_{Clayton}$ and $\hat{\rho}_{Gauss}$
- (e) Decide whether the gap in (c) is sampling error with an i.i.d. bootstrap of the weekly pairs (1999 draws, 95% percentile interval)

What to report and interpret

- A table with the four parameters, their tail dependence and the standard errors of (d)
- The gap of (c) with its bootstrap interval and a decision at the 5% level
- Interpretation: can $\hat{\tau}$ alone choose the copula family? Can the pair $(\hat{\tau}, \hat{\rho}_S)$ reject one?

A2: Kendall's Tau, Copula Parameters and Their Precision: Solution (1/2) [Solved]

Solution

- (a) Gaussian: $\rho = \sin(\pi \times 0.564/2) = 0.775$
- Clayton: $\theta = 2 \times 0.564/0.436 = 2.589$; Gumbel: $\theta = 1/0.436 = 2.295$; Frank: the numerical root is $\theta = 7.044$
- (b) Clayton: $\lambda_L = 2^{-1/2.589} = 0.765$; Gumbel: $\lambda_U = 2 - 2^{1/2.295} = 0.647$; Gaussian and Frank: no tail dependence
- The same $\hat{\tau}$ gives tail dependence between 0 and 0.77, depending only on the family chosen
- (c) $\frac{6}{\pi} \arcsin(0.775/2) = 0.760$ against the sample 0.746: a gap of 0.0134, small in absolute terms; (e) checks whether it is noise

A2: Kendall's Tau, Copula Parameters and Their Precision: Solution (2/2) [Solved]

Solution

- (d) $\hat{\tau}$ is a U-statistic with kernel h ; $E[h \mid Z_i] = 4C(U_i, V_i) - 2U_i - 2V_i + 1 = 2W_i + 1$, so Hoeffding gives $\text{Var}(\hat{\tau}) \approx 4\text{Var}(2W)/n = 16\text{Var}(W)/n$
- Substitution: $\text{se}(\hat{\tau}) = 4 \times 0.1239 / \sqrt{1393} = 0.0133$
- Clayton: $d\theta/d\tau = 2/(1-\tau)^2 = 10.53$, so $\text{se} = 10.53 \times 0.0133 = 0.140$; Gaussian: $d\rho/d\tau = \frac{\pi}{2} \cos(\pi\tau/2) = 0.993$, so $\text{se} = 0.0132$ (Genest & Rivest (1993))
- (e) Bootstrap: gap 0.0134, se 0.0026, 95% interval [0.0088, 0.0187]; zero is outside, so the gap is not sampling error at 5%
- Why so precise: $\hat{\tau}$ and $\hat{\rho}_S$ move together across samples, so their difference has a much smaller se than each of them
- Interpretation: $\hat{\tau}$ alone fits every family, but the pair $(\hat{\tau}, \hat{\rho}_S)$ already rejects the Gaussian copula for these data

A3: The Forbes–Rigobon Adjustment with an Estimated δ : Task (1/2) [Solved]

Context and data

- Model: $y_t = \alpha + \beta x_t + \varepsilon_t$ with β and σ_ε^2 constant; only the variance of the source market x changes in the crisis, $\sigma_{x,c}^2 = (1 + \delta)\sigma_x^2$
- Stylised numbers for (a)–(c): calm correlation 0.40, crisis correlation 0.60, source variance 4 times higher in the crisis
- Data for (d)–(e): 2-day returns of the S&P 500, Euro Stoxx 50 and BET, calm 14 Sep 2007 – 12 Sep 2008, crisis 15 Sep 2008 – 31 Mar 2009; $\hat{\delta} = 6.54$, effective $n_c = 65$ and $n_0 = 121$ (half the overlapping 2-day returns, an approximation)

What to do

- (a) Compute δ and the adjusted crisis correlation $\rho^* = \rho_c / \sqrt{1 + \delta(1 - \rho_c^2)}$
- (b) Compute the crisis correlation we would measure if the true correlation stayed at 0.40, using $\rho_c = \rho \sqrt{(1 + \delta)/(1 + \delta\rho^2)}$
- (c) Decide between contagion and interdependence

A3: The Forbes–Rigobon Adjustment with an Estimated δ : Task (2/2) [Solved]

What to do

- (d) Under i.i.d. Normality $(1 + \hat{\delta})/(1 + \delta) \sim F(n_c - 1, n_0 - 1)$; derive $\text{Var}(\hat{\delta})$ to first order
- (e) By the delta method in $(\hat{\rho}_c, \hat{\delta})$, including their covariance (both use the crisis window), compute $\text{se}(\rho^*)$ with δ fixed and with δ random, for Euro Stoxx 50 and BET in 2008

What to report and interpret

- The numbers of (a)–(b), each with the formula and the substituted values
- $\text{se}(\hat{\delta})$, the two standard errors of ρ^* and their percentage difference
- Interpretation: does treating δ as random change the 2008 verdict of no contagion?

A3: The Forbes–Rigobon Adjustment with an Estimated δ : Solution (1/2) [Solved]

Solution

- (a) Variance ratio 4, so $\delta = 4 - 1 = 3$
- $\rho^* = 0.60 / \sqrt{1 + 3(1 - 0.36)} = 0.60 / \sqrt{2.92} = 0.60 / 1.709 = 0.351$
- (b) $\rho_c = 0.40 \sqrt{(1 + 3) / (1 + 3 \times 0.16)} = 0.40 \sqrt{4 / 1.48} = 0.40 \times 1.644 = 0.658$
- (c) The adjusted correlation 0.351 is below the calm 0.40, and volatility alone would have lifted 0.40 to 0.658, more than the observed 0.60
- Conclusion: the whole rise is explained by higher volatility of the source market, i.e. interdependence, not contagion

A3: The Forbes–Rigobon Adjustment with an Estimated δ : Solution (2/2) [Solved]

Solution

- ▣ (d) $\text{Var}[\ln \chi_k^2/k] \approx 2/k$, so $\text{Var} \ln(1 + \hat{\delta}) \approx \frac{2}{n_c - 1} + \frac{2}{n_0 - 1}$ and $\text{Var}(\hat{\delta}) \approx 2(1 + \delta)^2 \left[\frac{1}{n_c - 1} + \frac{1}{n_0 - 1} \right]$
- ▣ Substitution: $\text{se}(\hat{\delta}) = 7.54 \sqrt{2(1/64 + 1/120)} = 1.65$
- ▣ (e) With $k = 1 + \delta(1 - \rho_c^2)$: $g_r = \partial \rho^* / \partial \rho_c = (1 + \delta)k^{-3/2}$, $g_\delta = \partial \rho^* / \partial \delta = -\rho_c(1 - \rho_c^2)k^{-3/2}/2$; $\text{Var}(\hat{\rho}_c) \approx (1 - \rho_c^2)^2/n_c$
- ▣ Bivariate Normal: $\text{Cov}(\hat{\rho}_c, \ln \hat{\sigma}_{x,c}^2) \approx \rho_c(1 - \rho_c^2)/n_c$, so $\text{Cov}(\hat{\rho}_c, \hat{\delta}) \approx (1 + \delta)\rho_c(1 - \rho_c^2)/n_c$;
 $\text{Var}(\rho^*) \approx g_r^2 \text{Var} \hat{\rho}_c + g_\delta^2 \text{Var} \hat{\delta} + 2g_r g_\delta \text{Cov}(\hat{\rho}_c, \hat{\delta})$
- ▣ Euro Stoxx 50 ($\rho_c = 0.804$): $g_r = 1.253$, $g_\delta = -0.0236$; se 0.055 with δ fixed, 0.051 with δ random (−7%)
- ▣ BET ($\rho_c = 0.485$): se $0.049 \rightarrow 0.047$ (−3%); testing $\rho^* - \hat{\rho}_{\text{calm}}$ also uses $\text{Cov}(\hat{\delta}, \hat{\rho}_{\text{calm}}) \approx -(1 + \delta)\rho_{\text{calm}}(1 - \rho_{\text{calm}}^2)/n_0$:
 $z = -0.19$ (BET), $z = -2.44$ (Euro Stoxx 50)
- ▣ Interpretation: both adjusted correlations stay below the calm ones, so the verdict of no contagion holds; a random δ need not widen the interval: a volatile crisis window raises $\hat{\delta}$ and $\hat{\rho}_c$ together, and their errors partly cancel in ρ^*

A4: Deriving the Forbes–Rigobon Bias: Task (1/2) [Proposed]

Solution discussed in class

Context and data

- Model $y_t = \alpha + \beta x_t + \varepsilon_t$ with $\text{Cov}(x_t, \varepsilon_t) = 0$; β and σ_ε^2 are the same in the calm period and in the crisis
- In the crisis only the variance of x changes: $\sigma_{x,c}^2 = (1 + \delta)\sigma_x^2$, $\delta > 0$

What to do

- (a) Write ρ^2 as a function of $k = \beta^2 \sigma_x^2 / \sigma_\varepsilon^2$
- (b) Derive the measured crisis correlation $\rho_c = \rho \sqrt{(1 + \delta) / (1 + \delta \rho^2)}$ and show that $|\rho_c| > |\rho|$ for $0 < |\rho| < 1$, i.e. $\rho_c > \rho$ when $\rho > 0$
- (c) Invert the relation to obtain $\rho^* = \rho_c / \sqrt{1 + \delta(1 - \rho_c^2)}$
- (d) State which assumption fails if a global shock raises both σ_x and σ_ε , and in which direction the adjustment is then biased

A4: Deriving the Forbes–Rigobon Bias: Task (2/2) [Proposed]

Solution discussed in class

What to report and interpret

- Each step of the derivation with one sentence of justification; the answer to (d) in economic terms
- Model: A3

A5: DCC Algebra: Task (1/2) [Proposed]

Solution discussed in class

Context and data

- DCC recursion $Q_t = (1 - a - b)\bar{Q} + a\varepsilon_{t-1}\varepsilon_{t-1}^\top + bQ_{t-1}$ with $a, b \geq 0$, $a + b < 1$, $\bar{Q}$ positive definite, $Q_0 = \bar{Q}$
- B1 estimates for SPY and TLT: $a = 0.0522$, $b = 0.9335$

What to do

- (a) Show by induction that Q_t is positive definite for every t
- (b) Assuming $\mathbb{E}[\varepsilon_t\varepsilon_t^\top] = \bar{Q}$ and stationarity, show that $\mathbb{E}[Q_t] = \bar{Q}$; explain why this assumption does not hold exactly for the standard DCC
- (c) Compute the approximate half-life $\ln 0.5 / \ln(a + b)$ of a shock to Q_t with the B1 estimates, say why it is only approximate for the standard DCC, and describe the case $a + b = 1$
- (d) For the corrected DCC of [Aielli \(2013\)](#), $Q_t = (1 - a - b)\bar{Q}^* + aQ_{t-1}^{*1/2}\varepsilon_{t-1}\varepsilon_{t-1}^\top Q_{t-1}^{*1/2} + bQ_{t-1}$ with $Q_t^* = \text{diag}(Q_t)$, show that $\mathbb{E}[Q_t] = \bar{Q}^*$ exactly

A5: DCC Algebra: Task (2/2) [Proposed]

Solution discussed in class

What to report and interpret

- The four short proofs and the half-life in days; interpretation: what does the half-life say about how fast a DCC hedge ratio drifts, and what else decides how often to rebalance?
- Model: B1

A6: Attainable Correlations: Task (1/2) [Proposed]

Solution discussed in class

Context and data

- ▣ $X = e^{\sigma_1 Z_1}$ and $Y = e^{\sigma_2 Z_2}$ with $Z_1, Z_2 \sim N(0, 1)$: the lognormal margins are fixed, the dependence between Z_1 and Z_2 is free
- ▣ Tools: Hoeffding's identity $\text{Cov}(X, Y) = \iint [F(x, y) - F_1(x)F_2(y)] dx dy$ and the Fréchet–Hoeffding bounds $W \leq C \leq M$

What to do

- ▣ (a) Show that $\rho(X, Y)$ is maximal for $Z_2 = Z_1$ (comonotone) and minimal for $Z_2 = -Z_1$ (countermonotone)
- ▣ (b) Derive $\rho_{\max}$ and $\rho_{\min}$ in closed form from the lognormal moments
- ▣ (c) Evaluate both for $\sigma_1 = 1$, $\sigma_2 = 2$

A6: Attainable Correlations: Task (2/2) [Proposed]

Solution discussed in class

What to report and interpret

- The two formulas and the two numbers; interpretation: what does this imply for reading a Pearson correlation of 0.3 between two price levels?
- Model: A2

B1: DCC for SPY and TLT: Task (1/2) [Solved]

Context and data

- Data: daily adjusted closes of SPY (SPY.US) and TLT (TLT.US) in data/market, prices joined on common days first, then log returns in %; 31 July 2002 – 18 September 2026, $T = 6\,073$
- Model: GARCH(1,1) with constant mean for each series, then DCC(1,1) with correlation targeting (Gaussian quasi-likelihood)

What to do

- (a) Fit GARCH(1,1) to each series and compute the standardised residuals $\varepsilon_{it} = (r_{it} - \hat{\mu}_i) / \hat{\sigma}_{it}$
- (b) Estimate (a, b) by maximising the correlation part of the likelihood; report Hessian standard errors and the likelihood ratio against CCC ($a = b = 0$), with its p -value from a parametric bootstrap of both steps under CCC (199 paths)
- (c) Re-estimate the two steps on 2002–2019 and on 2020–2026

B1: DCC for SPY and TLT: Task (2/2) [Solved]

What to do

- (d) Test for a change in the correlation of ε_t at an *unknown* date with Wied, Krämer & Dehling (2012) (Bartlett bandwidth $\lfloor T^{1/4} \rfloor$, 5% level) and date the break
- (e) On the same 199 residual-bootstrap paths, compute standard errors with step 1 held fixed (step 2 only) and with GARCH re-estimated on each path (both steps); compare both sets with the Hessian ones (Newey & McFadden (1994))

What to report and interpret

- A table of $(a, b, a + b, \bar{Q}_{12}, T)$ for the three samples; the likelihood ratio; the break statistic, its date and the two correlations; the three sets of standard errors
- Interpretation (1): is the persistence $a + b$ stable across subsamples, and what does a moving $\bar{Q}$ mean for a full-sample DCC?
- Interpretation (2): why is the χ^2 reference only indicative for the likelihood ratio, and why is a Fisher z test at January 2022, a date read off the lecture chart, not valid?

B1: DCC for SPY and TLT: Solution (1/3) [Solved]

Solution

- ▣ (a) GARCH(1,1), returns in %: SPY $\omega = 0.0283$, $\alpha = 0.127$, $\beta = 0.848$; TLT $\omega = 0.0057$, $\alpha = 0.049$, $\beta = 0.943$
- ▣ (b) Step 2: $\hat{a} = 0.0522$ (se 0.0059), $\hat{b} = 0.9335$ (se 0.0084), $\hat{a} + \hat{b} = 0.9857$, target $\bar{Q}_{12} = -0.248$
- ▣ Likelihood ratio: $2(\ell_C^{DCC} - \ell_C^{CCC}) = 2(460.3 - 193.5) = 533.4$
- ▣ χ_2^2 is not the null law: under $a = 0$ the parameter b is not identified (Davies problem) and the criterion is two-step; bootstrap under CCC (both steps, $B = 199$): largest null statistic 15.7, $p = 0.005$: constant correlation is rejected
- ▣ (c) 2002–2019: $a = 0.0565$, $b = 0.9199$, $\bar{Q}_{12} = -0.350$ ($T = 4\,386$); 2020–2026: $a = 0.0341$, $b = 0.9527$, $\bar{Q}_{12} = 0.003$ ($T = 1\,687$)
- ▣ Reading: persistence is similar ($a + b \approx 0.98$) but the target moves from about -0.35 to about 0; a full-sample DCC pulls the recent correlation towards a target that no longer applies

B1: DCC for SPY and TLT: Solution (2/3) [Solved]

Solution

- (d) Statistic: $Q_T = \max_j \frac{j}{\sqrt{T\bar{D}}} |\hat{\rho}_j - \hat{\rho}_T|$ with bandwidth $\lfloor 6.073^{1/4} \rfloor = 8$; result $Q_T = 4.25$
- p -value from the Kolmogorov distribution of $\sup |B|$: below 0.001; decision at 5% (1.358): reject constant correlation
- Estimated break 3 August 2020: residual correlation -0.361 before and 0.066 after; testing the two segments again gives a second break on 9 July 2007 ($Q = 3.01$) and $Q = 1.56$ ($p = 0.015$) after the main break
- The fixed-date Fisher test at January 2022 ($z = 14.9$) is not valid: the date was picked after seeing the data, and Fisher assumes i.i.d. Normal pairs
- Economic reading: the stock–bond relation changed in mid-2020, before the 2022 inflation shock, and it changed more than once

B1: DCC for SPY and TLT: Solution (3/3) [Solved]

Solution

- (e) Procedure: decorrelate the residuals with the Cholesky factor of R_t ; resample them; simulate DCC innovations with $(\hat{a}, \hat{b}, \bar{Q})$; (i) re-estimate step 2 only, (ii) rebuild returns with the fitted GARCH and re-estimate both steps
- Standard errors of $(\hat{a}, \hat{b})$: Hessian 0.0059, 0.0084; step 2 only 0.0051, 0.0077; both steps 0.0052, 0.0077
- How to read it: step 1 adds only 2% and 1% because $T \approx 6\,000$ pins down the GARCH parameters; the Hessian understates $se(\hat{b})$ by -8% because the innovations are not Normal
- Conclusion: the inference on (a, b) is robust here, but in short samples or with weak GARCH identification the two-step correction must be made

B2: Gaussian or t Copula for S&P 500 and Euro Stoxx 50?: Task (1/2) [Solved]

Context and data

- Data: the weekly returns and pseudo-observations of A2 (2000–2026, $n = 1\,393$); canonical maximum likelihood (CML) on the ranks
- The Gaussian copula is the t copula with $\nu = \infty$, a point on the boundary of the parameter space

What to do

- (a) Fit the Gaussian and the t copula by CML and compare them by AIC
- (b) Compute the likelihood ratio for $H_0 : \nu = \infty$ and its p -value by parametric bootstrap under H_0 (499 draws)
- (c) Run the Rosenblatt goodness-of-fit test ([Genest, Rémillard & Beaudoin \(2009\)](#)) for both copulas, parametric bootstrap with 499 draws, 5% level

B2: Gaussian or t Copula for S&P 500 and Euro Stoxx 50?: Task (2/2) [Solved]

What to do

- (d) Build 90% parametric-bootstrap intervals for ν and λ (200 draws) and compare $\hat{\lambda}$ with the empirical $\hat{C}(q, q)/q$ at $q = 5\%$
- (e) Compute the standard errors of $\hat{\rho}$ (and $\hat{\nu}$) from the inverse Hessian and from the [Genest, Ghoudi & Rivest \(1995\)](#) sandwich

What to report and interpret

- A table with $\hat{\rho}$, $\hat{\nu}$, ℓ , AIC and the fit p -value; the likelihood ratio with its bootstrap p ; the intervals of (d); the two sets of standard errors
- Interpretation: why not use the χ^2_1 distribution for the likelihood ratio? Does passing the fit test prove that the t copula is right? Why do the ranks add variance?

B2: Gaussian or t Copula for S&P 500 and Euro Stoxx 50?: Solution (1/2) [Solved]

Solution

	$\hat{\rho}$	$\hat{\nu}$	ℓ	AIC	fit p
Gaussian	0.771	—	623.3	-1244.6	0.002
Student t	0.775	3.91	674.2	-1344.4	0.874

- (a) The t copula has the lower AIC by 99.8 points: the extra parameter ν is strongly supported
- (b) Likelihood ratio $2(674.2 - 623.3) = 101.8$; bootstrap $p = 0.002$ from 499 draws under H_0 (the smallest possible is $1/500$); decision at 5%: reject the Gaussian copula
- χ^2_1 is wrong because $\nu = \infty$ lies on the boundary; the textbook fix is the mixture $\frac{1}{2}\chi^2_0 + \frac{1}{2}\chi^2_1$, the bootstrap avoids any such approximation
- (c) Fit test: Gaussian $p = 0.002$, rejected at 5%; t $p = 0.874$, not rejected; not rejecting is absence of evidence against the t copula, not proof

B2: Gaussian or t Copula for S&P 500 and Euro Stoxx 50?: Solution (2/2) [Solved]

Solution

- (d) $\hat{\nu} = 3.91$ with 90% interval $[2.98, 4.77]$; $\hat{\lambda} = 0.467$ with $[0.423, 0.518]$
- Empirical at $q = 5\%$: lower 0.517, upper 0.488; both near $\hat{\lambda}$, with a hint of asymmetry that the symmetric t copula ignores
- (e) Formula: sandwich $A^{-1}\Sigma A^{-1}/n$ with $\Sigma = \text{Var}[\dot{\ell}_{\theta} + W_1(U) + W_2(V)]$, against A^{-1}/n for the inverse Hessian
- Results: Gaussian $\text{se}(\hat{\rho})$ 0.0086 versus 0.0138; t $\text{se}(\hat{\rho})$ 0.0115 versus 0.0131; $\text{se}(\hat{\nu})$ 0.56 versus 0.58
- Interpretation: the ranks are estimated, so the Hessian understates the uncertainty of $\hat{\rho}$; ν is driven by the tails, where the ranking error matters little

B3: Clayton or Gumbel for Joint Losses of BET and Euro Stoxx 50?: Task (1/2) [Proposed]

Solution discussed in class

Context and data

- Data: daily closes of BET and the Euro Stoxx 50 (STOXX50E.INDX) from 5 January 2010, joined on common days (the two markets close almost at the same time); $T = 4\,154$
- Filter each series with GARCH(1,1) and work with **losses** $L = -\varepsilon$ and their pseudo-observations

What to do

- (a) Fit the Clayton, Gumbel, Gaussian and t copulas by CML; report AIC and the implied λ_L, λ_U
- (b) Run the Rosenblatt fit test with parametric bootstrap for each family (200 draws; 100 for the t copula) at the 5% level
- (c) Compute the empirical $\hat{\lambda}_U(q)$ and $\hat{\lambda}_L(q)$ of the losses at $q = 5\%$ and 10%

B3: Clayton or Gumbel for Joint Losses of BET and Euro Stoxx 50?: Task (2/2) [Proposed]

Solution discussed in class

What to report and interpret

- ▣ A table by family (parameter, AIC, λ_L , λ_U , fit p) and the four empirical tail coefficients
- ▣ Interpretation: in which corner of the loss copula are the joint crashes, why can Clayton on losses not capture them, and what would a rotated Clayton represent?
- ▣ Model: B2 (fit and test), A2 (parameters and tails)

B4: Forbes–Rigobon for 2008 and 2020: Task (1/2) [Proposed]

Solution discussed in class

Context and data

- Data: 2-day log returns $\ln P_t - \ln P_{t-2}$ of the S&P 500 (source), Euro Stoxx 50 and BET (targets), on common trading days
- Crises: 15 Sep 2008 – 31 Mar 2009 and 20 Feb – 30 Apr 2020; calm: the year before each crisis; effective n = half the number of overlapping 2-day returns (an approximation)

What to do

- (a) Compute ρ_{calm} , ρ_c , δ and ρ^* ; test $H_0 : \rho_c \leq \rho_{calm}$ and $H_0 : \rho^* \leq \rho_{calm}$ with one-sided Fisher z tests at 5%
- (b) Repeat with crisis windows shortened to 31 Dec 2008 and 31 Mar 2020
- (c) Treat δ as random: stationary bootstrap within each window (mean block 10 days, 1 999 draws); report the 90% interval of ρ^* and the bootstrap p for $H_0 : \rho^* \leq \rho_{calm}$

B4: Forbes–Rigobon for 2008 and 2020: Task (2/2) [Proposed]

Solution discussed in class

What to report and interpret

- ▣ A table by crisis and market with δ , ρ_{calm} , ρ_c , ρ_{raw} , ρ^* , ρ_{adj} , plus the bootstrap results
- ▣ Interpretation: where does the raw test find contagion and does it survive the adjustment? Why is δ so much larger in 2020? Which assumption is least credible for a pandemic?
- ▣ Model: A3

B5: Bitcoin and the S&P 500: DCC with Bootstrap Bands: Task (1/2) [Proposed]

Solution discussed in class

Context and data

- Data: daily closes of Bitcoin (BTC-USD.CC) and the S&P 500 (GSPC.INDX) on common trading days since 18 September 2014, $T = 3018$; weekend Bitcoin moves enter the Monday return

What to do

- (a) Fit GARCH(1,1) and DCC(1,1); compute the average conditional correlation before and after 1 January 2020
- (b) Residual bootstrap: decorrelate $\eta_t = L_t^{-1}\varepsilon_t$ (L_t the Cholesky factor of R_t), resample η_t , simulate a DCC with $(\hat{a}, \hat{b})$ and re-estimate; 200 draws; 90% intervals for a and b
- (c) Recompute ρ_t on the original data with each bootstrap (a, b) and draw the 90% band

B5: Bitcoin and the S&P 500: DCC with Bootstrap Bands: Task (2/2) [Proposed]

Solution discussed in class

What to report and interpret

- ▣ The estimates with intervals, the two averages, the share of days since 2020 with the lower band above zero, and the chart
- ▣ Interpretation: is the post-2020 correlation significantly positive? Why does a simple block bootstrap of ε_t bias $\hat{b}$ downwards? Which uncertainty does the band ignore, and how would you add the first-step GARCH uncertainty?
- ▣ Model: B1

B6: A Dynamic t Copula for S&P 500 and Euro Stoxx 50: Task (1/2) [Proposed]

Solution discussed in class

Context and data

- Data: the weekly pseudo-observations of A2 and B2 (2000–2026, $n = 1\,393$); the static t copula of B2 is the benchmark
- Model: GAS t copula (Creal, Koopman & Lucas (2013), unit scaling): $\rho_t = \tanh f_t$, $f_{t+1} = \omega + A s_t + B f_t$,
 $s_t = \partial \ln c(\hat{u}_t, \hat{v}_t; \rho_t, \nu) / \partial f_t$, $f_1 = \omega / (1 - B)$

What to do

- (a) Derive s_t from the t-copula density and estimate (ω, A, B, ν) by maximum likelihood
- (b) Compute the likelihood ratio against the static t copula, plot ρ_t and report its minimum and maximum with dates

B6: A Dynamic t Copula for S&P 500 and Euro Stoxx 50: Task (2/2) [Proposed]

Solution discussed in class

What to report and interpret

- ▣ The estimates, the likelihood ratio, the range of ρ_t and the chart
- ▣ Interpretation: does the dependence move, and when is it lowest? Why is χ^2 not the reference for this likelihood ratio? B2 did not reject the static t copula: is that a contradiction?
- ▣ Model: B2 (fit), lecture slide on the dynamic t copula

C1: Did Romanian and Euro-Area Equities Become More Integrated after 2020?: Task (1/2) [Proposed]

Solution discussed in class

Context and data

- Data: BET and the Euro Stoxx 50, weekly (Friday) returns on common trading days, 2010–2019 against 2020–2026; daily data for a robustness check
- 2020 brought COVID-19 and the FTSE Russell upgrade of Romania to Secondary Emerging ([FTSE Russell \(2019\)](#)): is a higher correlation integration or only the crises inside the sample?

What to do

- (a) Estimate the change in weekly correlation with a 95% stationary block-bootstrap interval ([Politis & Romano \(1994\)](#), mean block 8 weeks, 2000 draws)
- (b) Two separate robustness checks against the same baseline: (i) the Forbes–Rigobon adjustment for the change in Euro Stoxx 50 volatility on the full post-2020 sample; (ii) the unadjusted correlation without the weeks of 20 Feb – 30 Apr 2020
- (c) Compare Kendall's τ , the BET beta on the Euro Stoxx 50, the lower-tail dependence at $q = 10\%$ and a weekly DCC across the two periods

C1: Did Romanian and Euro-Area Equities Become More Integrated after 2020?: Task (2/2) [Proposed]

Solution discussed in class

What to report and interpret

- A one-slide table of all measures in both periods and a verdict with its caveats; why weekly and not daily returns?

C2: Critique an AI Answer [Proposed]

Solution discussed in class

The prompt sent to an artificial intelligence (AI) assistant

- “e holds the GARCH-standardised residuals of JPM and BAC (two US bank stocks). Calibrate a t copula with $\nu = 4$ from Kendall's τ and give me its tail dependence.”

The assistant's answer

```
import numpy as np
from scipy import stats
tau = stats.kendalltau(e['JPM'], e['BAC'])[0]
rho = 2 * np.sin(np.pi * tau / 6)
u = e.rank() / len(e)
z = stats.t.ppf(u, df=4)
nu = 4
x = np.sqrt((nu + 1) * (1 - rho) / (1 + rho))
lam = 2 * stats.t.cdf(-x, df=nu)

# Kendall's tau
# copula correlation
# pseudo-observations
# input of the t-copula likelihood

# tail dependence
```

- “With $\hat{\tau} = 0.50$: $\rho = 0.518$ and $\lambda_L = \lambda_U = 0.276$.”

C2: Critique an AI Answer, the Task [Proposed]

Solution discussed in class

- Context: the code above runs on the GARCH-standardised residuals of JPM and BAC (daily, 2010–2026), but the answer contains **three** mistakes typical of copula modelling; everything else is correct
 - ▶ Hint: what is u for the largest residual, and what does `stats.t.ppf` return there?
- (a) For each mistake, name the line and explain in one sentence why it is wrong
- (b) Say how you would detect it: a step-by-step check, a formula from A1 or A2, or a small simulation (state which)
- (c) Write the corrected line(s) and recompute ρ and λ for $\hat{\tau} = 0.50$, $\nu = 4$
- (d) Log each mistake in `AI_ERRORS.md`: what was asked, what the AI answered, what was wrong, how you found out, the correct version
- Report: the three corrected lines and the corrected numbers; interpretation: which of the three errors changes λ the most, and why?

C3: Energy Prices and EU Stock Markets, a Project Seed: Task (1/2) [Proposed]

Solution discussed in class

Context and data

- The case study of Lecture 6 ([Greco, Cramer, Pele & Lessmann \(2025\)](#)) estimates DCC-GARCH correlations between EU stock indices and Brent crude oil, 2010–2023: in every region the correlation peaks in 2020 and reaches its minimum in 2022 ([working paper \(2024\)](#), Table 1)
- Data: DAX (Western Europe) and WIG20, PX, BUX, BET (Central and Eastern Europe) from data/market; the daily Brent spot price (Europe) of the U.S. Energy Information Administration, via FRED; join prices on common trading days first, then compute log returns

What to do

- (a) For each index, estimate a bivariate DCC(1,1) model with GARCH(1,1) margins on the index–Brent pair over 2010–2023 and report a , b and $a + b$ with two-step standard errors, as in B1
- (b) Locate the maximum and the minimum of ρ_t with their dates, and check whether they fall in 2020 and 2022 and whether the DAX correlation exceeds the Central and Eastern European ones, as in the regional ranking of the case study

C3: Energy Prices and EU Stock Markets, a Project Seed: Task (2/2) [Proposed]

Solution discussed in class

What to do

- (c) Compare the mean DCC correlation of 24 February 2022 – 31 December 2023 with that of 2017–2019, using a 95% stationary block-bootstrap interval, and apply the Forbes–Rigobon adjustment for the change in Brent volatility, as in C1
- (d) Extend the sample to 2026, fit the Gaussian, t and Clayton copulas to the GARCH-standardised residuals before and after February 2022, and estimate the lower-tail dependence with bootstrap intervals, as in B2

What to report and interpret

- A table with a , b , $a + b$, the dates and values of the extreme correlations, the 2022–2023 change with its interval and the adjusted value, for each index
- Interpretation: is the post-2022 energy–equity link a volatility effect, a tail phenomenon or a change in ordinary co-movement? Which differences from the case study come from spot rather than futures prices?
- Model: B1 (DCC and two-step standard errors), C1 (bootstrap and Forbes–Rigobon), B2 (copulas and tail dependence)

Common Pitfalls

- ▣ Computing returns before joining series, or using daily correlations across time zones
- ▣ Testing a break at a date read off a chart; standard errors that ignore estimated ranks or the GARCH step
- ▣ Using χ^2 critical values for tests on the boundary ($a = 0$ in DCC, $\nu = \infty$ in the t copula)
- ▣ Fitting copulas to raw returns instead of GARCH-filtered pseudo-observations
- ▣ Treating a non-rejected copula as the true one, or a rejected one as useless: large samples detect small departures, and a rejection does not say which feature (e.g. time variation) is missing
- ▣ Block-bootstrapping a highly persistent process with short blocks: biased persistence and misleading bands

Take-Home Assignment (Team Project)

- ▣ Choose two markets not covered in class (e.g. WIG20 and BUX, gold and Bitcoin, two bank stocks)
- ▣ Estimate EWMA, rolling and DCC correlations; test DCC against CCC; show the correlation in two crises with the Forbes–Rigobon adjustment
- ▣ Fit five copulas to GARCH-filtered pseudo-observations; report AIC, bootstrap fit tests and tail dependence with intervals
- ▣ Log every AI error you catch in `AI_ERRORS.md` (at least three per project)
- ▣ Deliver a Quantlet folder (code, Metainfo, charts) and three slides: evidence, interpretation, limitations

Further Study on Quantinar

Advanced courses on quantinar.com

- ☐  Measuring Statistical Risk
- ☐  Multivariate Statistical Analysis

Code and data

- ☐ Seminar notebook (Colab); Quantlets: github.com/danpele/MFM/tree/main/Quantlets/Ch_06

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